Comparing Means

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Introduction

This page reviews the standard methods for comparing the mean of a continuous outcome between groups: t-tests, confidence intervals for a difference in means, and one-way analysis of variance.

This page builds on three others:

- Exploratory Data Analysis defines the sample statistics these methods use;
- Statistical Inference defines hypotheses, test statistics, p-values, confidence intervals, and the t, chi-square, and F reference distributions;
- Estimation defines estimators and their standard errors.

The t-tests and ANOVA on this page are the exact, Gaussian-outcome tests in the table of exact and approximate tests on the Maximum Likelihood page, which pairs each one with its large-sample counterpart.

This page is adapted from Vittinghoff et al. [1], Chapter 3.

The HERS data

The "heart and estrogen/progestin study" (HERS) was a clinical trial of hormone therapy for prevention of recurrent heart attacks and death among 2,763 post-menopausal women with existing coronary heart disease (CHD) [2].

The trial was conducted at 20 US clinical centers. Participants were randomized to receive either conjugated equine estrogens (0.625 mg/day) plus medroxyprogesterone acetate (2.5 mg/day) or a matching placebo [2]. Women were followed for an average of 4.1 years [2].

The primary outcome was nonfatal myocardial infarction or CHD death [2].

The HERS data are distributed with Vittinghoff et al. [1] on the book's companion website, as a Stata file that R can read directly:

```
# one unbroken string, so that link checkers test the whole URL:
url <- "https://regression.ucsf.edu/sites/g/files/tkssra16191/files/wysiwyg/home/data/hersda
hers <- haven::read_dta(url)
```

The `rmb` R package includes the same file, which these notes use so that rendering does not depend on the website. `haven::as_factor()` converts the Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
hers |> head()
#> # A tibble: 6 x 37
#>   HT      age raceth nonwhite smoking drinkany exercise physact globrat poorfair
#>   <fct> <dbl> <fct>  <fct>    <fct>   <fct>    <fct>    <fct>   <fct>   <fct>
#> 1 plac~    70 Afric~ yes      no      no       no       much m~ good    no
#> 2 plac~    62 Afric~ yes      no      no       no       much l~ good    no
#> 3 horm~    69 White  no       no      no       no       about ~ good    no
#> 4 plac~    64 White  no       yes     yes      no       much l~ good    no
#> 5 plac~    65 White  no       no      no       no       somewh~ good    no
#> 6 horm~    68 Afric~ yes      no      yes      no       about ~ good    no
#> # i 27 more variables: medcond <dbl>, htnmeds <fct>, statins <fct>,
#> #   diabetes <fct>, dmpills <fct>, insulin <fct>, weight <dbl>, BMI <dbl>,
#> #   waist <dbl>, WHR <dbl>, glucose <dbl>, weight1 <dbl>, BMI1 <dbl>,
#> #   waist1 <dbl>, WHR1 <dbl>, glucose1 <dbl>, tchol <dbl>, LDL <dbl>,
#> #   HDL <dbl>, TG <dbl>, tchol1 <dbl>, LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>,
#> #   SBP <dbl>, DBP <dbl>, age10 <dbl>
```

The examples on this page use these variables:

Variable	Meaning
`HT`	Randomized treatment: placebo or hormone therapy
`age`	Age at baseline (years)
`raceth`	Race/ethnicity: White, African American, or Other
`exercise`	Exercises at least three times per week (no/yes)
`BMI`	Body mass index at baseline (kg/m^2)
`SBP`	Systolic blood pressure at baseline (mmHg)
`glucose`	Fasting glucose at baseline (mg/dL)
`glucose1`	Fasting glucose at the year-1 visit (mg/dL)

```
hers |>
  dplyr::select(HT, age, raceth, exercise, BMI, SBP, glucose, glucose1) |>
  dplyr::glimpse()
#> Rows: 2,763
#> Columns: 8
#> $ HT       <fct> placebo, placebo, hormone therapy, placebo, placebo, hormone ~
#> $ age      <dbl> 70, 62, 69, 64, 65, 68, 70, 69, 61, 62, 72, 73, 52, 57, 57, 6~
#> $ raceth   <fct> African American, African American, White, White, White, Afri~
#> $ exercise <fct> no, no, no, no, no, no, no, yes, yes, no, no, no, no, no, yes~
#> $ BMI      <dbl> 23.69, 28.62, 42.51, 24.39, 21.90, 29.05, 34.45, 23.16, 30.26~
#> $ SBP      <dbl> 138, 118, 134, 152, 175, 174, 119, 178, 162, 111, 122, 158, 1~
#> $ glucose  <dbl> 84, 111, 114, 94, 101, 116, 120, 95, 105, 98, 111, 95, 97, 10~
#> $ glucose1 <dbl> 94, 78, 98, 93, 92, 115, NA, 95, 113, 98, 96, 107, 90, 100, 1~
```

Descriptive statistics

The Exploratory Data Analysis page defines the sample statistics used on this page, each with an example:

- the sample mean;
- the sample median;
- the sample variance;
- the sample standard deviation;
- the interquartile range;
- the sample proportion.

That page also defines the graphs used here, including box plots and scatter plots.

> **Example 0.1** (HERS baseline characteristics by treatment group)**.** Table 1 summarizes the HERS participants at baseline, separately for each randomized group. Continuous variables are summarized by sample mean (standard deviation), and categorical variables by count (percent).

Table 1: HERS: baseline characteristics by treatment group

```
hers |>
  dplyr::select(
    age, BMI, glucose, SBP, DBP,
    HT, raceth, exercise, smoking, diabetes
  ) |>
  gtsummary::tbl_summary(
    by = HT,
    statistic = list(
      gtsummary::all_continuous() ~ "{mean} ({sd})",
      gtsummary::all_categorical() ~ "{n} ({p}%)"
    ),
    digits = gtsummary::all_continuous() ~ 1,
    label = list(
      age ~ "Age (years)",
      BMI ~ "BMI (kg/m^2)",
      glucose ~ "Fasting glucose (mg/dL)",
      SBP ~ "Systolic BP (mmHg)",
      DBP ~ "Diastolic BP (mmHg)",
      raceth ~ "Race/ethnicity",
      exercise ~ "Exercises regularly",
      smoking ~ "Current smoker",
      diabetes ~ "Diabetes"
    )
  ) |>
  gtsummary::add_overall() |>
  gtsummary::bold_labels()
```

Characteristic	**Overall** N = 2,763[1]	**placebo** N = 1,383[1]	**hormone therapy** N =
Age (years)	66.6 (6.7)	66.8 (6.7)	66.5 (6.6)
BMI (kg/m^2)	28.6 (5.5)	28.5 (5.5)	28.6 (5.5)
Unknown	5	4	1
Fasting glucose (mg/dL)	112.2 (36.8)	112.4 (36.8)	111.9 (36.9)
Systolic BP (mmHg)	135.1 (19.0)	135.1 (19.4)	135.0 (18.7)
Diastolic BP (mmHg)	73.2 (9.7)	73.1 (9.7)	73.2 (9.7)
Unknown	1	1	0
Race/ethnicity			
White	2,451 (89%)	1,239 (90%)	1,212 (88%)
African American	218 (7.9%)	102 (7.4%)	116 (8.4%)
Other	94 (3.4%)	42 (3.0%)	52 (3.8%)
Exercises regularly	1,068 (39%)	530 (38%)	538 (39%)
Current smoker	360 (13%)	182 (13%)	178 (13%)
Diabetes	731 (26%)	352 (25%)	379 (27%)

[1]Mean (SD); n (%)

Because treatment was assigned at random, the two groups differ at baseline only by chance, and their summaries are close.

Comparing two groups: continuous outcomes

Hypotheses

The Statistical Inference page defines the null hypothesis, the alternative hypothesis, test statistics, p-values, and significance levels. In a comparison of two groups with means μ_1 and μ_2, the null hypothesis is usually $H_0 : \mu_1 = \mu_2$, and the two-sided alternative is $H_1 : \mu_1 \neq \mu_2$.

One-sample t-test

Definition 0.1 (One-sample t-test)**.** Let $x_1, \ldots, x_n$ be observations with $n \geq 2$, sample mean $\bar{x}$, and sample standard deviation s. The **one-sample t-test** of $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ uses the statistic

$$t \stackrel{\text{def}}{=} \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Its p-value is $\Pr(|T| \geq |t|)$, where T has the t_{n-1} distribution (t-distribution).
Source: Wikipedia contributors [3].

Theorem 0.1 (Null distribution of the one-sample t statistic)**.** *Let $X_1, \ldots, X_n$ be independent, each Gaussian with mean μ_0 and variance σ^2, with $n \geq 2$. Then the statistic $T \stackrel{def}{=} (\bar{X} - \mu_0)/(S/\sqrt{n})$ of Definition 0.1, computed from these random variables, has the t_{n-1} distribution.*

Proof

Proof. We use a standard fact about Gaussian samples [4, Sec. 5.5], pp. 203-205: $\bar{X}$ and S^2 are independent, and $V \stackrel{\text{def}}{=} (n-1)S^2/\sigma^2$ has the χ^2_{n-1} distribution. Also, $Z \stackrel{\text{def}}{=} (\bar{X} - \mu_0)/(\sigma/\sqrt{n})$ has the standard Gaussian distribution, and Z is independent of V because $\bar{X}$ is independent of S^2. So by the definition of the t-distribution, $Z/\sqrt{V/(n-1)}$ has the t_{n-1} distribution, and this ratio equals T:

$$
\begin{aligned}
\frac{Z}{\sqrt{V/(n-1)}} &= \frac{(\bar{X}-\mu_0)/(\sigma/\sqrt{n})}{\sqrt{S^2/\sigma^2}} && \text{(substitute } Z \text{ and } V\text{)} \\
&= \frac{(\bar{X}-\mu_0)/(\sigma/\sqrt{n})}{S/\sigma} && (\sqrt{S^2} = S, \text{ since } S \geq 0) \\
&= \frac{\bar{X}-\mu_0}{S/\sqrt{n}} && \text{(the factors of } \sigma \text{ cancel)} \\
&= T.
\end{aligned}
$$

□

Remark (Non-Gaussian data)

Remark 0.1 (Non-Gaussian data). When the observations are not Gaussian, T still has approximately the standard Gaussian distribution when n is large, by the central limit theorem, and then t_{n-1} is close to the standard Gaussian distribution (t quantiles approach Gaussian quantiles).

Paired t-test

Definition 0.2 (Paired t-test)**.** Let $(x_{1,1}, x_{1,2}), \ldots, (x_{n,1}, x_{n,2})$ be pairs of related measurements, such as the same participants measured at two times, and let $d_i \stackrel{\text{def}}{=} x_{i,2} - x_{i,1}$ be the within-pair differences, with population mean μ_d. The **paired t-test** of $H_0 : \mu_d = 0$ against $H_1 : \mu_d \neq 0$ is the one-sample t-test (Definition 0.1) of $H_0 : \mu = 0$ applied to $d_1, \ldots, d_n$.
Source: Wikipedia contributors [3].

Example 0.1 (Change in fasting glucose over the first year of HERS)**.** We test whether mean fasting glucose changed between baseline (`glucose`) and the year-1 visit (`glucose1`), across both treatment groups. Participants missing either measurement are dropped. The t statistic of Definition 0.1, computed on the differences:

```r
glucose_change <- hers |>
  dplyr::filter(!is.na(glucose), !is.na(glucose1)) |>
  dplyr::mutate(d = glucose1 - glucose) |>
  dplyr::pull(d)
n_pairs <- length(glucose_change)
t_paired <- mean(glucose_change) / (sd(glucose_change) / sqrt(n_pairs))
c(
  n = n_pairs,
  mean_change = mean(glucose_change),
  t = t_paired,
  p_value = 2 * pt(-abs(t_paired), df = n_pairs - 1)
)
#>           n mean_change           t     p_value
#> 2.61300e+03 2.62036e+00 4.15081e+00 3.41911e-05
```

`t.test()` with `paired = TRUE` gives the same statistic, along with a confidence interval for μ_d:

```r
t.test(hers$glucose1, hers$glucose, paired = TRUE)
#>
#>  Paired t-test
#>
#> data:  hers$glucose1 and hers$glucose
#> t = 4.151, df = 2612, p-value = 3.42e-05
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  1.38248 3.85824
#> sample estimates:
#> mean difference
#>         2.62036
```

Mean fasting glucose rose by about 2.6 mg/dL over the year, and the p-value is far below 0.05.

Two-sample t-tests

Definition 0.3 (Welch's two-sample t-test)**.** Let two independent groups have sample sizes $n_1, n_2 \geq 2$, sample means $\bar{x}_1$ and $\bar{x}_2$, and sample variances s_1^2 and s_2^2. **Welch's two-sample t-test** of $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$ uses the statistic

$$t \stackrel{\text{def}}{=} \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

and the Welch–Satterthwaite degrees of freedom

$$\hat{\text{df}} \stackrel{\text{def}}{=} \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

Its p-value is $\Pr(|T| \geq |t|)$, where T has the $t_{\hat{\text{df}}}$ distribution (t-distribution).
Source: Wikipedia contributors [5].

i Remark (Properties of Welch's test)

Remark 0.2 (Properties of Welch's test). Welch's test does not assume that the two groups have equal variances. Even for Gaussian data, $t_{\hat{\text{df}}}$ is only an approximation to the null distribution of t. For large groups, the central limit theorem makes the statistic approximately standard Gaussian under H_0, as in the WCGS cholesterol example. Welch's test is the default in R's `t.test()`.

Example 0.2 (Baseline fasting glucose by treatment group in HERS)**.** We test $H_0 : \mu_{\text{HT}} = \mu_{\text{placebo}}$ against $H_1 : \mu_{\text{HT}} \neq \mu_{\text{placebo}}$, where μ_{HT} and μ_{placebo} are the population mean baseline fasting glucose levels in the hormone therapy and placebo populations. Figure 1 compares the two groups' distributions.

```r
hers |>
  ggplot2::ggplot() +
  ggplot2::aes(x = HT, y = glucose) +
  ggplot2::geom_boxplot() +
  ggplot2::labs(
    x = "Treatment group",
    y = "Fasting glucose (mg/dL)"
  )
```

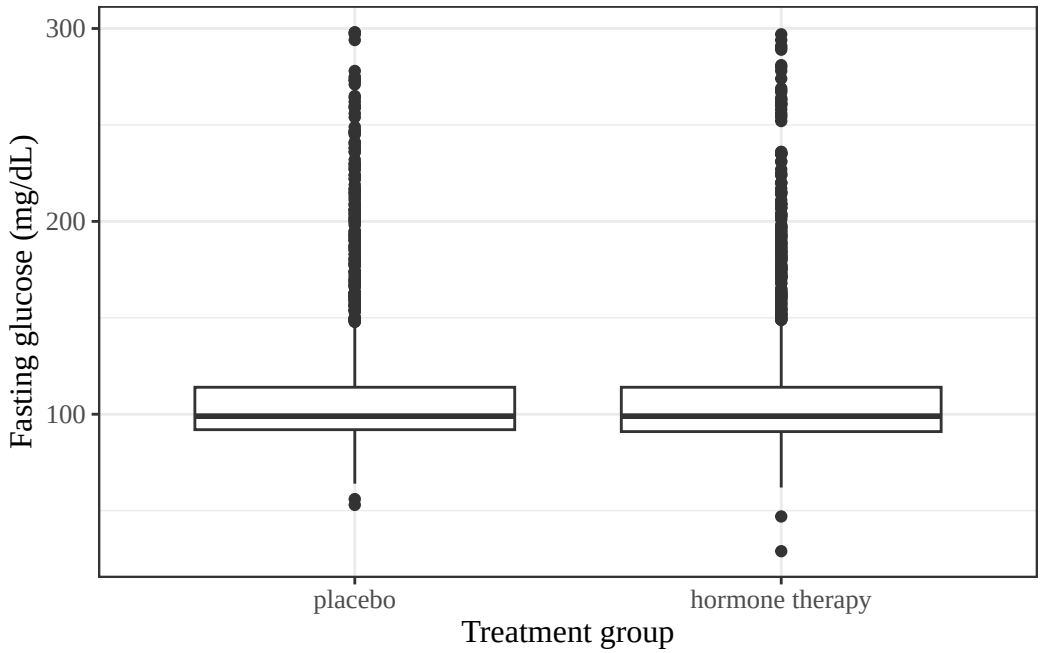


Figure 1: Baseline fasting glucose by treatment group in HERS

The statistic and degrees of freedom of Definition 0.3:

```r
glucose_ht <- hers |>
  dplyr::filter(HT == "hormone therapy") |>
  dplyr::pull(glucose)
glucose_placebo <- hers |>
  dplyr::filter(HT == "placebo") |>
  dplyr::pull(glucose)
v_ht <- var(glucose_ht) / length(glucose_ht)
v_placebo <- var(glucose_placebo) / length(glucose_placebo)
t_welch <- (mean(glucose_ht) - mean(glucose_placebo)) / sqrt(v_ht + v_placebo)
df_welch <- (v_ht + v_placebo)^2 /
  (v_ht^2 / (length(glucose_ht) - 1) +
     v_placebo^2 / (length(glucose_placebo) - 1))
c(t = t_welch, df = df_welch, p_value = 2 * pt(-abs(t_welch), df_welch))
#>           t          df     p_value
#>   -0.424594 2760.978546    0.671165
```

`t.test()` reports the same values:

```r
t.test(glucose_ht, glucose_placebo)
#>
#>  Welch Two Sample t-test
#>
#> data:  glucose_ht and glucose_placebo
#> t = -0.4246, df = 2761, p-value = 0.671
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  -3.34503  2.15423
#> sample estimates:
#> mean of x mean of y
#>   111.854   112.449
```

The p-value is large, so the data give no evidence against H_0. That result is expected here: treatment was randomized, so the groups' baseline sample means differ only by chance.

Definition 0.4 (Pooled variance)**.** For two samples of sizes n_1 and n_2 with sample variances s_1^2 and s_2^2, the **pooled variance** is

$$s_p^2 \stackrel{\text{def}}{=} \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

Source: Wikipedia contributors [6].

Example 0.3 (Pooled variance of two small samples)**.** With $n_1 = 3$, $s_1^2 = 4$, $n_2 = 5$, and $s_2^2 = 9$:

$$\begin{aligned} s_p^2 &= \frac{(3-1)\cdot 4 + (5-1)\cdot 9}{3+5-2} && \text{(definition of pooled variance)} \\ &= \frac{8+36}{6} && \text{(arithmetic)} \\ &\approx 7.33 && \text{(arithmetic)} \end{aligned}$$

The pooled variance lies between the two sample variances, closer to s_2^2, whose sample is larger.

Remark (What the pooled variance estimates)

Remark 0.3 (What the pooled variance estimates)*.* When the two populations share one variance σ^2, the pooled variance estimates it.

Definition 0.5 (Pooled two-sample t-test)**.** With the notation of Definition 0.3 and the pooled variance s_p^2, the **pooled two-sample t-test** of $H_0 : \mu_1 = \mu_2$ uses the statistic

$$t_p \overset{\text{def}}{=} \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}.$$

Its p-value is $\Pr(|T| \geq |t_p|)$, where T has the $t_{n_1+n_2-2}$ distribution.
Source: Wikipedia contributors [3].

Theorem 0.2 (Null distribution of the pooled t statistic)**.** *Let the observations in both groups be independent and Gaussian, all with the same mean and the same variance σ^2. Then t_p (Definition 0.5), computed from these random variables, has the $t_{n_1+n_2-2}$ distribution [4, Sec. 8.2], p. 371.*

Remark (The equal-variance assumption)

Remark 0.4 (The equal-variance assumption)*.* Theorem 0.2 needs equal variances in the two groups, and Welch's test (Definition 0.3) does not, so these notes use Welch's test by default.

Example 0.4 (Pooled t-test of baseline glucose in HERS)**.** In HERS, the two groups have nearly equal sizes (1380 and 1383) and nearly equal standard deviations (36.9 and 36.8

mg/dL), so the pooled test gives almost the same result as Welch's test in Example 0.2:

```
t.test(glucose_ht, glucose_placebo, var.equal = TRUE)
#>
#>  Two Sample t-test
#>
#> data:  glucose_ht and glucose_placebo
#> t = -0.4246, df = 2761, p-value = 0.671
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  -3.34502  2.15422
#> sample estimates:
#> mean of x mean of y
#>   111.854   112.449
```

Read

- Hogg et al. [4, Sec. 8.2], pp. 369–377 covers tests of the equality of two means.
- Rice [7, Sec. 11.2.1], pp. 421–432 covers normal-theory methods for two independent samples.
- Rice [7, Sec. 11.3.1], pp. 446–447 covers normal-theory methods for paired samples.

Confidence intervals for the difference in means

Definition 0.6 (Welch confidence interval for a difference in means)**.** With the notation of Definition 0.3, the **Welch** $100(1-\alpha)\%$ **confidence interval** for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\hat{\mathrm{df}},\, 1-\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

where $t_{\hat{\mathrm{df}},\, 1-\alpha/2}$ is the $1-\alpha/2$ quantile of the $t_{\hat{\mathrm{df}}}$ distribution. It is an approximate confidence interval.

Example 0.5 (Confidence interval for the HERS baseline glucose difference)**.** Continuing Example 0.2, the 95% interval of Definition 0.6 is:

```
(mean(glucose_ht) - mean(glucose_placebo)) +
  c(lower = -1, upper = 1) * qt(0.975, df_welch) * sqrt(v_ht + v_placebo)
#>    lower    upper
#> -3.34503  2.15423
```

This interval matches the one `t.test()` reports in Example 0.2. It contains 0, consistent with the large p-value there.

Read

Hogg et al. [4, Sec. 7.2], pp. 314–322 covers confidence intervals for the difference of two means.

One-way analysis of variance

Definition 0.1 (Between-group sum of squares)**.** Let $y_{j1}, \ldots, y_{jn_j}$ be the n_j observations in group j, for $k \geq 2$ groups with $n \stackrel{\text{def}}{=} \sum_{j=1}^{k} n_j$ observations in all, let $\bar{y}_j$ be the sample mean of group j, and let $\bar{y}$ be the sample mean of all n observations. The **between-group sum of squares** is

$$\text{SS}_{\text{between}} \stackrel{\text{def}}{=} \sum_{j=1}^{k} n_j (\bar{y}_j - \bar{y})^2.$$

Source: Wikipedia contributors [8].

Example 0.1 (Between-group sum of squares of two small groups)**.** Let group 1 be $\{1, 2, 3\}$ and group 2 be $\{4, 5, 6\}$, so $\bar{y}_1 = 2$, $\bar{y}_2 = 5$, and $\bar{y} = 3.5$. Then:

$$\begin{aligned} \text{SS}_{\text{between}} &= 3(2 - 3.5)^2 + 3(5 - 3.5)^2 && \text{(definition)} \\ &= 3(2.25) + 3(2.25) && \text{(square the deviations)} \\ &= 13.5 && \text{(arithmetic)} \end{aligned}$$

Definition 0.2 (Within-group sum of squares)**.** With the notation of Definition 0.1, the **within-group sum of squares** is

$$\text{SS}_{\text{within}} \stackrel{\text{def}}{=} \sum_{j=1}^{k} \sum_{i=1}^{n_j} (y_{ji} - \bar{y}_j)^2.$$

Source: Wikipedia contributors [8].

Example 0.2 (Within-group sum of squares of two small groups)**.** For the groups of Example 0.1:

$$\begin{aligned}
\mathrm{SS}_{\text{within}} &= [(1-2)^2 + (2-2)^2 + (3-2)^2] + [(4-5)^2 + (5-5)^2 + (6-5)^2] && \text{(definition)} \\
&= 2 + 2 && \text{(arithmetic)} \\
&= 4 && \text{(arithmetic)}
\end{aligned}$$

Definition 0.3 (Mean squares)**.** In one-way analysis of variance, the **mean squares** are the sums of squares divided by their degrees of freedom: $\mathrm{MS}_{\text{between}} \overset{\text{def}}{=} \mathrm{SS}_{\text{between}}/(k-1)$ and $\mathrm{MS}_{\text{within}} \overset{\text{def}}{=} \mathrm{SS}_{\text{within}}/(n-k)$. When the k groups share one variance σ^2, $\mathrm{MS}_{\text{within}}$ estimates it.

Example 0.3 (Mean squares of two small groups)**.** For the groups of Example 0.1, $k = 2$ and $n = 6$, so

$$\begin{aligned}
\mathrm{MS}_{\text{between}} &= 13.5/1 \\
&= 13.5
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{MS}_{\text{within}} &= 4/4 \\
&= 1.
\end{aligned}$$

Definition 0.4 (One-way analysis of variance)**.** With the mean squares of $k \geq 2$ groups and $n > k$ observations in all, the **one-way analysis of variance (ANOVA)** F-test of

$$\begin{aligned}
H_0 : \mu_1 &= \mu_2 \\
&= \cdots \\
&= \mu_k
\end{aligned}$$

against the alternative that at least two group means differ uses the statistic [9]

$$F \overset{\text{def}}{=} \frac{\mathrm{MS}_{\text{between}}}{\mathrm{MS}_{\text{within}}}.$$

Its p-value is $\Pr(F^* \geq F)$, where F^* has the $F_{k-1,\,n-k}$ distribution (F-distribution). Source: Wikipedia contributors [10].

ℹ Remark (Interpreting the F statistic)

Remark 0.1 (Interpreting the F statistic). Large values of F mean that the group means are spread out more than the variation within groups would explain. For the groups of

Example 0.3,

$$\begin{aligned} F &= 13.5/1 \\ &= 13.5. \end{aligned}$$

Theorem 0.1 (Null distribution of the ANOVA F statistic)**.** *Let all n observations be independent, with observation y_{ji} Gaussian with mean μ_j and the same variance σ^2 in every group. If*

$$\begin{aligned} H_0 : \mu_1 &= \cdots \\ &= \mu_k \end{aligned}$$

holds, then F (Definition 0.4), computed from these random variables, has the $F_{k-1,\,n-k}$ distribution [4, Sec. 9.3], p. 449.

Example 0.4 (Fasting glucose by race/ethnicity in HERS)**.** The group sizes, sample means, and sample standard deviations of baseline fasting glucose:

```
hers |>
  dplyr::summarize(
    .by = raceth,
    n = dplyr::n(),
    mean = mean(glucose),
    sd = sd(glucose)
  )
#> # A tibble: 3 x 4
#>   raceth               n  mean    sd
#>   <fct>            <int> <dbl> <dbl>
#> 1 African American   218  123.  44.3
#> 2 White             2451  111.  35.6
#> 3 Other               94  125.  42.6
```

The F statistic of Definition 0.4, computed step by step:

```r
anova_parts <- hers |>
  dplyr::mutate(grand_mean = mean(glucose)) |>
  dplyr::mutate(.by = raceth, group_mean = mean(glucose)) |>
  dplyr::summarize(
    ss_between = sum((group_mean - grand_mean)^2),
    ss_within = sum((glucose - group_mean)^2),
    k = dplyr::n_distinct(raceth),
    n = dplyr::n()
  ) |>
  dplyr::mutate(
    f = (ss_between / (k - 1)) / (ss_within / (n - k)),
    p_value = pf(f, k - 1, n - k, lower.tail = FALSE)
  )
anova_parts
#> # A tibble: 1 x 6
#>   ss_between ss_within     k     n     f      p_value
#>        <dbl>     <dbl> <int> <int> <dbl>        <dbl>
#> 1     45919.  3704543.     3  2763  17.1 0.0000000414
```

$\text{SS}_{\text{between}}$ is summed here over observations rather than groups: each observation in group j contributes $(\bar{y}_j - \bar{y})^2$, which gives the n_j weights of Definition 0.4. `aov()` reports the same sums of squares, F statistic, and p-value:

```r
aov(glucose ~ raceth, data = hers) |>
  parameters::parameters() |>
  parameters::print_md(digits = 3)
```

Parameter	Sum_Squares	df	Mean_Square	F	p
raceth	45918.814	2	22959.407	17.105	< .001
Residuals	3.705e+06	2760	1342.226		

Anova Table (Type 1 tests)

The p-value is far below 0.05: population mean fasting glucose differs among the three race/ethnicity groups.

Remark (Unequal variances in the HERS glucose groups)

Remark 0.2 (Unequal variances in the HERS glucose groups). The group standard deviations differ, from 36 to 44 mg/dL, so the equal-variance condition of Theorem 0.1 is

questionable. `oneway.test()` performs Welch's version of the F-test, which does not assume equal variances, and reaches the same conclusion:

```
oneway.test(glucose ~ raceth, data = hers)
#>
#>  One-way analysis of means (not assuming equal variances)
#>
#> data:  glucose and raceth
#> F = 12.49, num df = 2.0, denom df = 185.7, p-value = 8.17e-06
```

One-way ANOVA is a special case of linear regression: it is the F-test comparing a linear regression model with a single categorical predictor to the model with an intercept only (Linear Models Overview). `lm()` gives the same F statistic as `aov()`:

```
lm(glucose ~ raceth, data = hers) |>
  anova() |>
  parameters::parameters() |>
  parameters::print_md(digits = 3, footer = "")
```

Parameter	Sum_Squares	df	Mean_Square	F	p
raceth	45918.814	2	22959.407	17.105	$< .001$
Residuals	3.705e+06	2760	1342.226		

With $k = 2$ groups, the ANOVA F statistic equals the square of the pooled t statistic (Definition 0.5), as the two treatment groups of Example 0.4 show:

```
c(
  f = anova(lm(glucose ~ HT, data = hers))[["F value"]][[1]],
  t_squared =
    t.test(glucose_ht, glucose_placebo, var.equal = TRUE)$statistic[[1]]^2
)
#>         f t_squared
#>  0.180281  0.180281
```

Read

- Hogg et al. [4, Sec. 9.3], pp. 446–455 covers one-factor analysis of variance.
- Casella and Berger [11, Sec. 11.2], pp. 522–538 covers oneway analysis of variance.
- Rice [7, Sec. 12.2], pp. 477–488 covers the one-way layout.

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